Tsunami Surrogate Modeling with Neural Operators
Fast Operator Learning for Synthetic Tsunami Wave Propagation

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Abstract

We study neural-operator surrogates for tsunami wave propagation using a synthetic data pipeline built from two-dimensional shallow-water simulations. The model input consists of bathymetry and an initial disturbance field, while the target is the resulting wave-height evolution over time. Our primary surrogate is a Fourier Neural Operator (FNO), compared against convolutional baselines under matched training and evaluation protocols. We design experiments around three questions: accuracy relative to the simulator, runtime speedup, and generalization to unseen bathymetry and source families. We additionally include an uncertainty analysis and a curriculum-by-resolution training strategy to test whether progressive scaling from coarse to fine grids improves high-resolution performance. The paper positions neural operators as promising fast surrogates for tsunami forecasting research, while clearly separating research-prototype claims from operational early-warning deployment.

1 Introduction

Physics-based tsunami simulation is computationally expensive when repeated many times across event scenarios, parameter sweeps, or inverse problems. This motivates the search for surrogate models that can approximate tsunami propagation much faster while preserving acceptable accuracy.

In this work, we frame surrogate modeling as operator learning: given a bathymetry field and an initial disturbance, predict the wave-height field across future timesteps. Fourier Neural Operators (FNOs) are a natural candidate because they learn mappings between functions and have shown strong performance on PDE-driven tasks [1].

Our goal is not to claim an operational warning system. Instead, we build a clean research benchmark that allows us to ask a sharper question: can a neural operator provide a meaningful speed-accuracy tradeoff for tsunami-like wave propagation under controlled synthetic conditions?

The main contributions of this study are:

1. a compact but reproducible synthetic tsunami data pipeline built from a shallow-water solver,
2. a neural-operator benchmark centered on FNOs with CNN, U-Net, and ConvLSTM baselines,
3. a paper-aligned evaluation suite covering accuracy, speed, generalization, and uncertainty, and
4. a curriculum-by-resolution experiment that tests whether progressive training improves high-resolution prediction quality.

2 Related Work

Neural operators. Neural operators extend learning from finite-dimensional vector mappings to mappings between functions. The Fourier Neural Operator introduced an efficient spectral parameterization that achieved strong PDE-surrogate results and demonstrated resolution transfer and zero-shot super-resolution behavior on benchmark problems [1].

Tsunami and shallow-water simulation. Long-wave propagation is commonly modeled with the shallow-water equations, often with finite-volume or related conservative discretizations that balance robustness and computational efficiency [3, 4]. For fault-driven seafloor deformation, Okada-style analytical surface deformation models remain a standard conceptual reference for synthetic uplift generation [2].

Toward real-time forecasting. Recent work on digital twins and probabilistic tsunami forecasting highlights the importance of fast forward prediction, inverse inference, and uncertainty quantification in realistic warning settings [5, 6]. Our repository is best understood as a research stepping stone toward that broader direction rather than a substitute for full-physics operational systems.

3 Methods

3.1 Problem formulation

We learn a surrogate map

$$\mathcal{G} : (b(x, y), s(x, y)) \mapsto \eta(x, y, t_{1:T}), \tag{1}$$

where b is bathymetry, s is the initial disturbance, and η is the wave-height evolution over T timesteps.

3.2 Synthetic data generation

Each sample is generated by:

1. sampling a random bathymetry field,
2. sampling a disturbance field from a source-family distribution,
3. evolving the shallow-water system forward in time, and
4. storing the input fields and the resulting wave-height trajectory.

The solver implementation is intentionally lightweight and serves as a consistent data generator rather than a high-fidelity regional tsunami model.

3.3 Surrogate models

Our main model is the FNO. We compare it with CNN, U-Net, and ConvLSTM baselines to test whether the operator-learning inductive bias improves accuracy and transfer.

3.4 Training objective

The training loss is a weighted combination of reconstruction and auxiliary penalties:

$$\mathcal{L} = \lambda_{\text{mse}}\mathcal{L}_{\text{mse}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}} + \lambda_{\text{mass}}\mathcal{L}_{\text{mass}}. \tag{2}$$

The exact enabled terms are determined by the experiment configuration.

3.5 Curriculum by resolution

A key extension in this project is curriculum-by-resolution training. The model is first trained on coarse grids and then continued on finer grids. This tests whether progressive resolution scaling improves optimization stability or final high-resolution performance.

4 Experimental Setup

4.1 Datasets and splits

We use synthetic train, validation, and test splits generated from the simulator. Unless otherwise stated, the train split is used to compute normalization statistics and the test split is reserved for final reporting.

4.2 Evaluation protocol

We report:

- reconstruction error (MSE, RMSE, MAE),
- runtime speedup relative to the simulator,
- generalization performance on unseen bathymetry and source families, and
- uncertainty diagnostics for dropout or ensemble-based prediction.

4.3 Experiment suite

The codebase is organized around the following experiment groups:

1. same-resolution learning,
2. unseen bathymetry generalization,
3. cross-resolution transfer,
4. physics-loss ablation,
5. uncertainty analysis,
6. curriculum by resolution.

4.4 Implementation details

This repository is implemented in Python using NumPy, SciPy, Matplotlib, PyYAML, and PyTorch. Training, evaluation, visualization, and dataset generation are all configuration-driven.

5 Results

5.1 Accuracy against the simulator

[TODO: Insert quantitative table for FNO vs baselines on the held-out test set.]

Table 1: Placeholder accuracy table. Replace with final numbers.

Model	MSE ↓	RMSE ↓	MAE ↓
CNN	–	–	–
U-Net	–	–	–
ConvLSTM	–	–	–
FNO	–	–	–

5.2 Speedup

[TODO: Insert wall-clock comparison between the trained surrogate and the numerical simulator.]

5.3 Generalization to unseen bathymetry

[TODO: Insert held-out bathymetry results and representative visualizations.]

5.4 Uncertainty analysis

[TODO: Show predictive spread vs empirical error for MC dropout or ensembles.]

5.5 Curriculum-by-resolution study

[TODO: Compare high-resolution training from scratch against progressive 32-¿64-¿128 training.]

6 Discussion

This project is designed as a controlled research benchmark. That has two consequences. First, the synthetic setting makes ablations and comparisons clean. Second, the conclusions should be interpreted at the level of surrogate-modeling methodology, not operational tsunami deployment.

The most interesting scientific question is likely not whether a single trained model can produce visually plausible waves, but whether neural operators preserve useful structure under distribution shift: new bathymetry, new source families, and higher-resolution domains. The curriculum experiment is particularly important here because it connects the empirical behavior of the repository directly to one of the strongest promises of neural operators.

A second promising direction is to move from forward surrogate modeling toward sensor-to-wave inversion, where sparse observations are used to infer the source and then forecast the wave field. That would align the project more closely with current probabilistic tsunami forecasting research.

7 Conclusion

We presented a clean code and manuscript scaffold for studying neural-operator surrogates for tsunami wave propagation. The repository combines a synthetic shallow-water simulator, FNO-centered surrogate modeling, and paper-ready evaluation for accuracy, speed, generalization, uncertainty, and curriculum-by-resolution transfer. The main value of the project is not only in baseline performance, but in providing a reproducible platform for stronger experiments and clearer scientific claims.

A Additional Implementation Notes

A.1 Repository mapping

The manuscript maps cleanly onto the codebase:

- simulator: `src/solver/`
- data generation: `src/data_gen/`
- models: `src/models/`
- training: `src/training/`
- evaluation: `src/evaluation/`
- figures: `results/` and `paper/figs/`

A.2 Suggested appendix content

Potential appendix material includes:

- hyperparameter tables,
- additional qualitative predictions,
- source-family definitions,
- solver details and numerical stability notes,
- failure-case examples.

References

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